Math 5310 Midterm 1 Key

1) $n=1$ $\quad (1+a)(a-1) = a^2-1 \Rightarrow (1+a) = \dfrac{a^2-1}{a-1}$

$n \Rightarrow n+1$ $\quad 1+a+\cdots+a^{n+1} = 1+a(1+a+\cdots a^n) = 1+a\,\dfrac{(a^{n+1}-1)}{a-1} = \dfrac{a-1+a^{n+2}-a}{a-1} = \dfrac{a^{n+2}-1}{a-1}$

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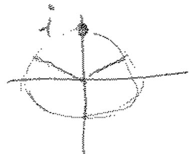
2) $z^3 = i$



$z_1 = \cos 30^\circ + i\sin 30^\circ = \frac{\sqrt{3}}{2} + \frac{i}{2}$

$z_2 = \cos 150^\circ + i\sin 150^\circ = -\frac{\sqrt{3}}{2} + \frac{i}{2}$

$z_3 = \cos 270 + i\sin 270 = -i$

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3) $\begin{cases} (a,b)=1 \Rightarrow 1 = ma+nb \\ a \mid bc \Rightarrow bc = a\cdot k \end{cases}$

so $c = mac+nbc = mac+nak = a(mc+nk) \Rightarrow a \mid c$

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4) $\{s_1, s_2, s_3\} \subset S$ $\quad$ Let $f(s_1)=s_2$, $f(s_2)=s_3$, $f(s_3)=s_1$ & $f(s)=s \;\forall s \notin \{s_1,s_2,s_3\}$

Let $g(s_1)=s_2$, $g(s_2)=s_1$ $\quad g(s)=s \;\forall s \notin \{s_1, s_2\}$ then $fg \neq gf$

In fact $fg(s_1) = f(s_2) = s_3 \quad gf(s_1) = g(s_2) = s_1$

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5) Let $V = \{m-qn \mid q \in \mathbb{Z},\; m-qn \geq 0\}$

By the WOP there is a smallest element $r = m-qn \in V$

Since $r \geq 0$, we must show $r<n$. If not, then

$0 \leq r-n = m-(q+1)n \in V$ But $r-n<r$ is a contradiction

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6) $m(S_n) = n!$

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7) 2^n

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1) 15 pts $\quad$ 2) 10 pts $\quad$ 3) 18 pts $\quad$ 4) 17 pts $\quad$ 5) 20 pts $\quad$ 6) 10 pts $\quad$ 7) 10 pts

total 100 pts